

Grushome: Accelerating development and adoption of visual IoT solutions for diverse usage scenarios

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

[About Grushome](#)



Grus Home
南京鹤梦信息技术有限公司

Grushome leverages industry-leading capabilities and a network of Google significantly reduce time to market of IoT software while improving its service and enhancing the

About Grushome

Founded in 2015, Grushome is specialized in visual IoT technologies that enable both enterprise and consumer applications. Through its one-stop visual IoT platform, Grushome not only develops software solutions for enterprise customers, but also offers easy-to-use SDKs to facilitate IoT application development. It currently has 106 enterprise customers providing IoT services to 35 million users on 80 million end devices around the world.

Industries: Technology

effectiveness of i
infrastructure.

Location: China

Google Cloud results

- Enables custom ML model training for visual IoT solutions with Vertex AI
- Ensures glitchless visual IoT services across regions through VPC
- Helps save 20% of operational costs with custom VMs in Compute Engine

Shorte
powered IoT
solution
develop
time by
months

Compute Engine

Google Cloud (<https://cloud.google.com/>)
Dataflow (<https://cloud.google.com/dataflow/>),

BigQuery (<https://cloud.google.com/bigquery/>),

Virtual Private Cloud

Compute Engine

(<https://cloud.google.com/compute/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Virtual Private Cloud

(<https://cloud.google.com/vpc/>)

- **Supports
cost-
effective
data
storage
and
processing
with
BigQuery
and
Dataflow**

Alongside the advancement of artificial intelligence (AI), visual Internet of Things (IoT) technologies, powered by AI-based image recognition, are increasingly adopted by organizations and households. From smart doorbells, smart cameras to smart city infrastructures, visual IoT solutions are facilitating security, business management, and urban governance around the world.

Since 2017, Grushome has been dedicated to expanding the application of visual IoT solutions with its expertise in visual IoT technologies. The company develops enterprise and consumer IoT products, as well as software development kits (SDKs) that can be used to swiftly develop visual IoT software for diverse usage scenarios ranging from home security, store management to factory operations. Through partnering with hardware manufacturers and telecom operators around the world, Grushome currently provides visual IoT services to 35 million users on 80 million end devices.

"To provide high-quality and innovative visual IoT services, having excellent global network connectivity and powerful ML model training resources is key. Google Cloud met all our needs in those aspects."

—Xiaodong Zeng, CEO,
Grushome

"Our visual IoT solutions are developer-friendly and highly integrated with cloud tools," explains Xiaodong Zeng, Chief Executive Officer at Grushome. "By leveraging our SDKs, developers can easily create visual IoT software that meets their specific needs and be ready to deploy in the cloud within one week. It would have taken them several months to build the same product from scratch."

To offer services to users around the world in compliance with local data security regulations, Grushome has been deploying its one-stop visual IoT platform that supports all its IoT solutions in several network regions. However, the public cloud platform that it first adopted had segregated regional networks, which undermined the quality of cross-region connectivity. Data transfer between different regions could be delayed up to three seconds, resulting in poor user experience. Since it was not possible to move backup systems from other regions when the infrastructure in one region crashed in a timely manner, the Grushome team needed to always have extra resources on hand to ensure smooth disaster recovery.

With that in mind, Grushome needed a cloud platform that offers better global connectivity to improve its IoT services. The company also wanted to replace its self-built machine learning (ML) model training system with fully managed tools for more cost-efficient visual IoT feature development. The company then chose to adopt Google Cloud (<https://cloud.google.com/?hl=en>), because it provides a single virtual network across all regions and AI capabilities that supports swift ML model training for diverse purposes.

"To provide high-quality and innovative visual IoT services, having excellent global network connectivity and powerful ML model training resources is key. Google Cloud met all our needs in those aspects," notes Zeng.

Accelerating visual IoT feature development with Vertex AI

In late 2022, Grushome migrated its one-stop visual IoT platform and development workloads to Google Cloud within four weeks. Now, the company uses Vertex AI (<https://cloud.google.com/vertex-ai>) to train its ML models during the feature development process. Before adopting Vertex AI, Grushome employed an on-prem ML training system, with which its development team needed to spend two months setting up a training framework for each newly added AI chip, and three to five days adapting the trained models to the cloud environment where they would eventually be deployed for testing. As Vertex AI offers fully managed ML tools that already run in the cloud, the team could immediately start the model training process without worrying about hardware settings and test the trained models directly on Google Cloud. This significantly reduced its time to market.

"Using Vertex AI
to develop visual
IoT features has
significantly
reduced the time
and effort
required for our

development
work. We can
leverage the fully
managed
platform and
many
sophisticated
pretrained
models to build
features for any
usage scenarios
within very little
time."

—Xiaodong Zeng, CEO,
Grushome

Another advantage of developing visual IoT features on Vertex AI is greater development flexibility. Now, Grushome and its enterprise customers can not only leverage a wide range of pretrained algorithms to quickly develop new visual IoT software, but also customize training parameters to fit specific usage scenarios. For example, Grushome partnered with one of its customers to build a wildlife identification feature in around two weeks by linking the animal image recognition algorithm offered by Vertex AI with its animal information database.

"Using Vertex AI to develop visual IoT features has significantly reduced the time and effort required for our development work. We can leverage the fully managed platform and many sophisticated pretrained models to build features for any usage scenarios within very little time," Zeng says. "It's like we're growing our business on the shoulders of giants."

Realizing glitchless cross-region connectivity with VPC

In terms of computing and data processing, moving to Google Cloud has also helped Grushome save considerable costs. The company uses the virtual machines (VMs) in [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>) to run its IoT services, some of which require high memory capacity but a small number of CPUs, while others demand a lot of CPU computing resources. Since the specifications of the VMs in Compute Engine can be customized, Grushome is able to set up VMs that meet the needs of its different services and reach a higher resource utilization rate that leads to 20% less costs.

On top of that, by setting up mirrored sources in Compute Engine, Grushome is able to use deployment scripts to realize one-click deployment for its visual IoT Platform-as-a-Service (PaaS), and provide basic service kits and SDKs for secondary development to global visual IoT developers. Zeng points out that this way, Grushome's customers can quickly build their own IoT platform software.

To manage the IoT data of its services, Grushome leverages BigQuery (<https://cloud.google.com/bigquery>) to store business and log data collected from IoT end devices, and Dataflow (<https://cloud.google.com/dataflow>) to tag and structuralize data. The processed data is then sent to its self-built IoT data analytics system deployed in Compute Engine for cross analyses and data management. Before the migration to Google Cloud, Grushome used an on-prem system to store and process its data, which required capacity expansion every three to four months as the company's business grew. For a larger capacity increase, the whole data storage architecture needed to be upgraded, and it would take three engineers one month to complete the work. Since BigQuery and Dataflow are fully managed and autoscalable, the Grushome team now can easily store and process any amount of data with minimum maintenance effort.

Expanding visual IoT features with ever-evolving AI technologies

Moving forward, Grushome will leverage the ML capabilities of Vertex AI to develop more innovative visual IoT services, such as a customer traffic analytics feature for retail stores and a baby security alert system. The company also plans to use Translation AI (<https://cloud.google.com/translate>) and the generative AI tools of Google to build customer service chatbots in different languages. To make its IoT products accessible to more customers, it will offer its SDKs and SaaS services on Google Cloud Marketplace

(<https://pantheon.corp.google.com/marketplace?pli=1>) for easy purchase and deployment.

"With its industry-leading AI capabilities and high-performance infrastructure, Google Cloud has helped us greatly improve our DevOps efficiency and service quality. As AI continues to advance, we're excited to explore more application potential of the visual IoT technologies together with Google Cloud."

—Xiaodong Zeng, CEO,
Grushome

Zeng says, "With its industry-leading AI capabilities and high-performance infrastructure, Google Cloud has helped us greatly improve our DevOps efficiency and service quality. As AI continues to advance, we're excited to explore more application potential of the visual IoT technologies together with Google Cloud."